CLINICAL · APPROACH TO

# Approach to a mouth ulcer

Most mouth ulcers are traumatic or aphthous and heal within two weeks. The approach is built around one rule: **a single ulcer that has not healed within three weeks must be biopsied** to exclude squamous cell carcinoma. Beyond that, the pattern of the ulcers, single or multiple, recurrent or persistent, with or without blisters, skin, eye, gut or genital disease, points to the cause, and simple blood tests find the treatable deficiencies behind many recurrent ulcers.

AT A GLANCE

- Single, persistent over 3 weeks, indurated: biopsy.
- Recurrent and healing: aphthous, check blood count, ferritin, B12, folate, coeliac.
- Multiple with skin, eye or genital lesions: systemic disease.
- Remove trauma and review in 2 weeks.

## 1 CHAPTER ONE Patterns

Single or multiple, acute or recurrent.

### 1.1 Sorting ulcers

MOUTH ULCERS BY PATTERN

| PATTERN | THINK OF | CLUES |
|---|---|---|
| Single, persistent over three weeks | Squamous cell carcinoma; traumatic ulcer; tuberculosis; syphilis | Induration, rolled edge, fixed; lateral tongue or floor of mouth; risk factors |
| Single, with a clear cause | Traumatic ulcer | Sharp tooth, denture, bite; heals within two weeks once the cause is removed |
| Recurrent, multiple, healing | Recurrent aphthous stomatitis | Onset in youth; minor, major or herpetiform |
| Acute, multiple, with fever | Viral infection | Primary herpes, hand, foot and mouth disease, herpangina |

| PATTERN | THINK OF | CLUES |
|---|---|---|
| Recurrent with genital ulcers and eye disease | Behçet's disease | Uveitis, skin lesions, pathergy |
| With gut symptoms or anaemia | Crohn's disease, coeliac disease, haematinic deficiency | Diarrhoea, weight loss, low ferritin, B12 or folate |
| Blisters, erosions and desquamative gingivitis | Pemphigus vulgaris, mucous membrane pemphigoid, lichen planus | Nikolsky sign, skin or eye involvement, white striae |
| Target lesions and crusted lips | Erythema multiforme | Recent herpes or drug |
| Drug-related | Nicorandil, NSAIDs, methotrexate, chemotherapy, bisphosphonates | Timing with the drug |
| Immunosuppressed | Neutropenia, HIV, CMV, fungal | Systemic disease or treatment |

DO NOT MISS

- **A single ulcer over three weeks,** especially indurated, with a raised edge, on the lateral tongue or floor of mouth, or with a neck node: urgent biopsy for squamous cell carcinoma.
- **Leukoplakia or erythroplakia** next to an ulcer.
- **Widespread erosions with skin blisters:** pemphigus vulgaris — urgent dermatology.
- **Ulcers with eye involvement:** pemphigoid or Behçet's disease — ophthalmology.
- **Ulcers with sore throat and fever in a patient on carbimazole or clozapine:** neutropenia.

# 2 CHAPTER TWO History and examination

The whole patient.

## 2.1 History

Ask how long each ulcer lasts and how often they come, the age of onset, the number and size, any trigger (trauma, stress, menstruation, foods, stopping smoking), and whether they scar. Ask about genital ulcers, eye symptoms, rashes, joint pain, bowel symptoms, weight loss, fatigue and fevers, and take a full drug history. Ask about smoking, alcohol, betel quid and HIV risk.

## 2.2 Examination

Note the site, size, number, base and edge of each ulcer, and feel it: induration and fixity suggest malignancy. Look for a sharp tooth or ill-fitting denture, white or red patches, blisters, gingival involvement and lichenoid striae. Examine the neck, skin, eyes and, when relevant, genitalia.

# 3 CHAPTER THREE Key causes

Recognising each.

## 3.1 Recurrent aphthous stomatitis

This is the commonest cause of recurrent ulcers, beginning in childhood or adolescence. **Minor** aphthae are under 1 cm, on non-keratinised mucosa, and heal in 10 to 14 days without scarring. **Major** aphthae are larger, deeper and last weeks, healing with scarring. **Herpetiform** ulcers are crops of many pinhead ulcers that coalesce. Triggers include trauma, stress, hormonal changes, sodium lauryl sulphate toothpastes, and deficiencies of iron, vitamin B12 and folate. Patients with recurrent ulcers should have a full blood count, ferritin, B12, folate and coeliac serology.



**FIGURE 1** Multiple aphthous ulcers on the uvula and faucial pillars.
Dhingra, fig. 43.1

## 3.2 Malignancy

Oral squamous cell carcinoma presents as a non-healing ulcer, often painless at first, with induration and a raised everted edge. Risk factors are smoking, alcohol, betel quid and chewing tobacco. Biopsy is taken from the edge, including normal tissue. The tongue and the floor of mouth are the highest-risk sites.



**FIGURE 2** First, a traumatic ulcerated mass of the tongue from sharp teeth. Second, a carcinoma to compare.
Dhingra, figs 43.2 and 43.3

## 3.3 Mucocutaneous disease

**Lichen planus** produces white lacy striae with erosions, often bilateral on the buccal mucosa; erosive lichen planus carries a small risk of malignant change and needs follow-up. **Pemphigus vulgaris** produces fragile blisters that rupture into painful erosions, often starting in the mouth; **mucous membrane pemphigoid** produces tougher blisters and desquamative gingivitis and can scar the eyes and larynx. Both need biopsy of perilesional tissue for **direct immunofluorescence**, as well as routine histology.

## 3.4 Systemic disease

**Behçet's disease** combines recurrent oral ulcers with genital ulcers, uveitis and skin lesions. Crohn's disease, coeliac disease, SLE, reactive arthritis, cyclic neutropenia and the PFAPA syndrome in children (periodic fever, aphthous stomatitis, pharyngitis and adenitis) are other causes.



**FIGURE 3** Palatal ulceration in a patient with a white cell count of 2,500 per cubic millimetre during chemotherapy.
Dhingra, fig. 43.4

# 4 CHAPTER FOUR Investigation and management

Test, biopsy, treat.

## 4.1 Investigation

Blood tests for recurrent ulcers include a full blood count, ferritin, B12, folate, and coeliac serology; inflammatory markers and an HIV test when indicated. Any persistent single ulcer is biopsied, and perilesional tissue is sent fresh for immunofluorescence when a blistering disease is suspected. Swabs for herpes simplex PCR help in acute vesicular ulcers.

## 4.2 Treatment

The cause is treated: a sharp tooth smoothed, a denture adjusted, a deficiency replaced, a causative drug stopped with the prescriber. Aphthous ulcers are managed with chlorhexidine or benzydamine mouthwashes, topical corticosteroids (such as betamethasone dissolved as a mouthwash, or beclometasone spray), avoiding triggers including sodium lauryl sulphate, and colchicine or other systemic agents in severe cases under specialist care. Blistering diseases, Behçet's disease and lichen planus are managed with oral medicine, dermatology and rheumatology colleagues.

THE MOUTH ULCER

1. **Single and persistent over three weeks?**
   - → **yes**
     urgent biopsy — exclude squamous cell carcinoma
   - → **single with obvious trauma**
     remove the cause, review in two weeks, biopsy if not healed
2. **Recurrent and multiple?**
   - → **bloods**
     full blood count, ferritin, B12, folate, coeliac serology

→ **minor or major aphthae**
topical treatment

→ **genital ulcers or eye disease**
Behçet's disease

3 Blisters, erosions or desquamative gingivitis?

→ biopsy with immunofluorescence

→ pemphigus, pemphigoid, lichen planus — specialist care

4 Acute with fever

→ herpes or other viral infection — supportive care, antiviral if severe

→ **on carbimazole or clozapine**
blood count

# 5 CHAPTER FIVE For the examination

What the examiner wants to hear.

## 5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **The three-week rule** and biopsy from the edge.
- **Recurrent aphthous stomatitis:** types and blood tests.
- **Pemphigus versus pemphigoid:** level of the split, immunofluorescence.
- **Behçet's disease** and other systemic causes.
- **Drug-induced ulceration:** nicorandil.

## 5.2 Common pitfalls

COMMON PITFALLS

- Giving repeated topical steroids for a single persistent ulcer that is a cancer.
- Biopsying only the necrotic centre of an ulcer.
- Missing iron, B12, folate or coeliac disease behind recurrent ulcers.

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Written for this app from Oxford Textbook of Otolaryngology ch. 54, Oxford Specialist Handbook ch. 3, Oxford Handbook of ENT ch. 3, Dhingra ch. 47. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

# On imaging

*The imaging of this topic: key findings, differentials and representative scans.*

ORAL CAVITY & TONGUE · AFTER SURGERY

## Squamous cell carcinoma: oral cavity, oropharynx and larynx

### IMAGING FINDINGS

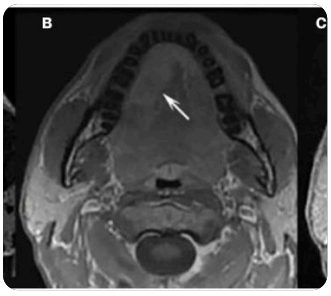
- Enhancing mucosal mass; depth of invasion in the tongue
- Crossing the lingual septum; neurovascular bundles
- Mandible cortex (CT) and marrow (MRI)

### TOP DIFFERENTIALS

- Lingual tonsil hyperplasia
- Abscess
- Minor salivary gland tumour

### CHECKLIST

- MRI for oral and oropharyngeal tumours
- Nodes on both sides; chest CT




Axial MRI of an oral cavity SCC shows a solid mass involving the left lateral border of the tongue showing intermediate signal intensity on T2…



Coronal T1WI MR in a woman with a painful right tongue ulcer shows subtle low signal intensity at the lateral tongue margin. There is no evidence …

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 504.